

VSB ENGINEERING COLLEGE

Karur

INTERNSHIP / INDUSTRY WITH PLACEMENT OPPORTUNITY

PROJECT REPORT

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Academic Year	2026 – 2027
section	ECE-D
Technology Stack	React • Supabase • PostgreSQL • Netlify

BONAFIDE CERTIFICATE

This is to certify that the project report entitled "**Personal Finance Tracker Web Application**" is the bonafide work of **Praveenkumar R** (Register Number: **922525106234**), who carried out the internship work under our supervision. This report is submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Engineering in electronics and communication and Engineering** at **VSB Engineering College, Karur**, during the academic year **2026–2027**

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ACKNOWLEDGEMENT

I thank **God Almighty** for the strength and wisdom to complete this internship successfully.

I am deeply grateful to the **Management of VSB Engineering College, Karur** for providing excellent infrastructure and academic support throughout my programme.

I extend my sincere thanks to the **Head of the Department of Computer Science and Engineering** for constant encouragement and administrative support.

I am profoundly grateful to my **Internal Guide** for invaluable guidance, timely suggestions, and constructive feedback at every stage of this project.

I also thank all the **Faculty Members** of the Department of CSE for their support and encouragement throughout the internship period.

Special appreciation goes to the **open-source communities** behind React, Supabase, PostgreSQL, and Netlify whose documentation and resources were instrumental in building this application.

Finally, I owe my deepest gratitude to my **parents and family** for their unconditional love, moral support, and patience.

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ABSTRACT

Personal financial management remains one of the most critical yet underserved challenges for individuals in the digital age. This internship project presents the design and development of a **Personal Finance Tracker Web Application** – a modern, full-stack solution built using React, Supabase, PostgreSQL, and deployed on Netlify.

The application provides an intuitive interface for users to register securely, log income and expenses across customisable categories, set monthly budgets, and gain insights through interactive visual dashboards. The backend leverages Supabase's PostgreSQL database with Row-Level Security (RLS) policies to ensure each user's financial data remains private and protected.

Key features include real-time transaction updates via Supabase WebSocket subscriptions, responsive React UI with Recharts visualisations, CSV export functionality, and a Progressive Web App (PWA) configuration for mobile access. The serverless architecture eliminates dedicated backend infrastructure, reducing operational costs and complexity.

This report details the complete software development lifecycle – from requirements analysis and literature review through database design, system architecture, implementation, and performance testing. Results confirm the application meets all defined functional and non-functional requirements with Lighthouse scores above 90 and sub-200 ms end-to-end operation times.

Keywords: Personal Finance, Web Application, React, Supabase, PostgreSQL, Netlify, Budget Tracker, Expense Management, Serverless Architecture, Full-Stack Development.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
CSS	Cascading Style Sheets
HTML	HyperText Markup Language
HTTPS	HyperText Transfer Protocol Secure
JS	JavaScript
JSON	JavaScript Object Notation
JWT	JSON Web Token
MFA	Multi-Factor Authentication
NPM	Node Package Manager
PWA	Progressive Web Application
RLS	Row-Level Security
REST	Representational State Transfer
SPA	Single Page Application
SQL	Structured Query Language
SSL	Secure Sockets Layer
TLS	Transport Layer Security
UI	User Interface
UX	User Experience
URL	Uniform Resource Locator
WCAG	Web Content Accessibility Guidelines
WAL	Write-Ahead Log
CDN	Content Delivery Network
BaaS	Backend as a Service

Chapter 1

Introduction

1.1 Background and Motivation

In the contemporary digital era, effective management of personal finances has become indispensable. With the rise of cashless transactions, subscription-based services, and e-commerce, individuals accumulate numerous financial interactions daily. Yet a significant majority of people lack the tools or habits to systematically track their income and expenditures, resulting in poor savings rates and financial stress.

Research by financial institutions consistently highlights that individuals who actively monitor their budgets demonstrate significantly better savings behaviour. The motivation for this project stems from the observation that while several finance applications exist, most are either (a) feature-heavy desktop software inaccessible on mobile, (b) paid subscription services with steep learning curves, or (c) simple spreadsheet templates that offer no interactivity or visual feedback.

This internship project bridges that gap – building a web-based, responsive, serverless personal finance tracker accessible from any browser, free to use, and backed by a robust cloud database with enterprise-grade security.

1.2 Problem Statement

The core problem addressed is the absence of a lightweight, intuitive, and secure web application that enables individuals to:

- Record daily income and expenditure transactions with categorisation.
- Set and monitor monthly budgets across different spending categories.
- Visualise financial patterns through charts and interactive summaries.
- Access financial data securely from any device without installing software.
- Export reports as CSV for offline use or sharing with financial advisors.

1.3 Objectives

- Design and implement a full-stack web application using React, Supabase, PostgreSQL, and Netlify.
- Implement secure user authentication with JWT-based session management via Supabase Auth.

-
- Design a normalised PostgreSQL schema with Row-Level Security to protect individual user data.
 - Build a responsive React frontend with reusable components and real-time state management.
 - Integrate Supabase real-time subscriptions so the UI updates instantly on data changes.
 - Deploy the application on Netlify with CI/CD integration for seamless continuous delivery.
 - Validate the system against functional and non-functional requirements through structured testing.

1.4 Scope of the Project

The scope encompasses the complete development lifecycle – from requirements analysis and architectural design through coding, integration, testing, and deployment. The application targets individual users managing personal or household finances and is not intended as an enterprise accounting solution. Future extensions may include multi-currency support, bank account integration via open banking APIs, and AI-driven spending predictions.

Chapter 2

Literature Review

A thorough review of existing literature and commercial products was conducted prior to the design phase. The review spans academic research on personal finance behaviour, technology papers on serverless web architectures, and an analysis of comparable applications.

2.1 Personal Finance Management – Behavioural Research

Thaler and Sunstein (2008) demonstrated through Nudge Theory that presenting financial information visually in digestible summaries significantly improves saving behaviour. Studies in the Journal of Financial Planning (2020) indicate that digital budgeting tool users save approximately 18% more per month than non-users, establishing a clear empirical basis for building tools that make financial data visible and actionable.

Lusardi and Mitchell (2014) further highlighted that financial literacy is strongly correlated with better retirement savings outcomes. Digital tools providing real-time spending feedback serve as proxies for financial literacy improvement, reinforcing the social value of the proposed application.

2.2 Evolution of Web Application Architectures

Traditional multi-tier web applications consisted of a presentation layer (HTML/CSS), an application server layer (PHP, Java EE, ASP.NET), and a relational database layer. While robust, this architecture imposed significant operational overhead in server management, scaling, and maintenance.

The introduction of REST APIs and later GraphQL enabled frontend decoupling from backend logic. React (Facebook, 2013) popularised the component-based SPA paradigm with efficient DOM diffing through a virtual DOM. The Backend-as-a-Service (BaaS) model, typified by Firebase and later Supabase (2020), democratised backend development by abstracting server management, authentication, realtime subscriptions, and storage into managed cloud services.

2.3 Review of Existing Finance Applications

Application	Key Features	Limitations
-------------	--------------	-------------

Mint (Intuit)	Budgeting, bank sync, credit score	US-only, in-app advertising, data privacy concerns
YNAB	Zero-based budgeting, goal tracking	Paid subscription (\$14.99/month)
Money Manager	Expense logs, charts, categories	Mobile-only, no web dashboard
Walnut	SMS parsing, Indian bank support	Limited customisation, inactive development
Spendee	Multi-currency, cloud sync	Premium tier restricts most useful features
GnuCash	Full double-entry accounting	Desktop-only, steep learning curve

2.4 Serverless and Jamstack Architecture

The Jamstack (JavaScript, APIs, Markup) architecture recommends pre-rendering static assets and consuming APIs at runtime. This yields faster page loads, lower infrastructure costs, and reduced attack surfaces compared to server-rendered monoliths. Netlify's continuous deployment pipeline automates build and deployment from Git repositories. Combined with Supabase's managed PostgreSQL, this stack enables developers to deliver production-grade applications without managing any server infrastructure.

2.5 Security Considerations in Financial Applications

Financial applications process sensitive personal data and must implement robust security measures including TLS/SSL encryption for data in transit, bcrypt-hashed password storage, JWT-based stateless authentication, and database-level access control. Supabase implements PostgreSQL Row-Level Security natively, ensuring SQL queries automatically filter results to the authenticated user's own records, eliminating cross-user data leakage even if application logic contains bugs.

Chapter 3

Existing System

3.1 Analysis of Existing Tools

The existing landscape of personal finance tools falls into three categories: desktop software (Quicken, GnuCash), mobile applications (Walnut, Money Manager), and web-based SaaS platforms (Mint, YNAB, Spendee).

Desktop Software: Applications like Quicken offer comprehensive accounting features but require local installation, are platform-specific, and do not sync across devices. Complex interfaces deter casual users needing only basic income/expense tracking.

Mobile-Only Applications: Convenient for on-the-go entry but lack desktop-sized dashboards for in-depth analysis. Data is often siloed on the device and cloud backup may require paid upgrades.

Web-Based Platforms: Mint offers automatic bank transaction sync but is restricted to US accounts and displays in-app advertising. YNAB employs a zero-based budgeting philosophy that, while effective, requires commitment and is gated behind a subscription fee.

3.2 Limitations of Existing Systems

- Geographic restrictions – Mint is limited to North America; Walnut supports only select Indian banks.
- Paid subscription models making premium features inaccessible to students and low-income users.
- Excessive permissions and privacy concerns through automatic bank account linking via third-party aggregators.
- Poor mobile responsiveness in web platforms and absence of desktop views in mobile apps.
- Limited customisation of categories, currencies, and report formats.
- Absence of real-time collaboration for household or family budgeting.
- High complexity with features (investment tracking, credit scores) irrelevant to basic users.
- Vendor lock-in with restricted data export options and poor interoperability.

Proposed System

4.1 System Overview

The proposed Personal Finance Tracker is a web-based, serverless application that directly addresses the identified limitations. It is designed to be universal (any browser, no installation), secure (PostgreSQL RLS), free (Netlify and Supabase free tiers), customisable (user-defined categories), visual (real-time Recharts dashboards), and exportable (CSV download).

The application follows the **Jamstack architecture**: the React frontend is statically pre-built and served from Netlify's global CDN, while all dynamic operations are delegated to Supabase's API layer backed by a managed PostgreSQL instance.

4.2 Advantages Over Existing Systems

Advantage	Detail
No geographic restriction	Manual entry works anywhere; no regional bank API required.
Zero cost	Free tier covers all features for individual users indefinitely.
Privacy-first	No advertising; user data is never shared with third parties.
Real-time updates	Supabase WebSocket subscriptions push DB changes to UI instantly.
Fully responsive	CSS Grid and Flexbox ensure usability on all screen sizes.
Open-source stack	All underlying technologies are open-source and independently auditable.
CSV data export	Users retain full ownership and portability of their financial data.
Serverless scaling	Netlify CDN scales automatically; zero server maintenance required.

Chapter 5

Software Requirements

5.1 Frontend: HTML, CSS, JavaScript / React

The frontend is implemented as a React 18 SPA using Vite as the build tool. Functional components with Hooks (useState, useEffect, useContext, useReducer) handle all state management without Redux or external stores.

React 18: Core UI library; concurrent rendering with automatic batching for improved performance.

React Router v6: Client-side routing for Login, Dashboard, Transactions, Budget, and Settings pages without full page reloads.

CSS Modules: Scoped CSS prevents style collisions; BEM naming conventions throughout.

Recharts: Declarative charting library built on D3 for BarChart, PieChart, and LineChart with responsive containers.

React Hook Form: Performant form validation with minimal re-renders for transaction and budget entry forms.

date-fns: Lightweight date utility for formatting, comparison, and range operations.

PapaParse: CSV parsing and generation for the data export feature.

Vite: Next-generation build tool offering instant HMR and optimised production bundles.

5.2 Backend: Supabase

Supabase provides the complete backend layer without a dedicated server. Key capabilities used:

- **Authentication:** Email/password signup and login with JWT session tokens; magic link (passwordless) login supported.
- **Database API:** Auto-generated RESTful API over PostgreSQL tables via the PostgREST middleware.
- **Realtime:** WebSocket subscriptions using PostgreSQL logical replication (WAL) to stream row changes to clients.
- **Row-Level Security:** SQL policies that automatically scope all data access to the authenticated user's UUID.

-
- **Storage:** Object storage for potential future receipt image uploads.
 - **Edge Functions:** Deno-based serverless functions for custom logic such as scheduled budget alert emails.

5.3 Database: PostgreSQL

PostgreSQL 15, managed by Supabase, serves as the relational database engine. It was selected for its ACID compliance, rich data types (including JSONB), full-text search, native UUID support, and mature extension ecosystem. The schema is designed in Third Normal Form (3NF) to minimise redundancy with referential integrity enforced via foreign key constraints.

5.4 Hosting: Netlify

- **Continuous Deployment:** Git-triggered builds from the main branch of the linked GitHub repository.
- **Global CDN:** Static assets distributed across 200+ edge nodes for sub-100 ms Time to First Byte globally.
- **HTTPS by default:** Automatic SSL/TLS certificate provisioning via Let's Encrypt at no additional cost.
- **Environment Variables:** Supabase URL and anon key stored as Netlify build-time variables; never exposed in version control.
- **SPA Redirects:** `_redirects` file configured so all paths serve `index.html`, enabling client-side routing.

5.5 Browser Requirements

Browser	Minimum Version	Notes
Google Chrome	90+	Recommended; full PWA and WebSocket support
Mozilla Firefox	88+	Full compatibility
Microsoft Edge	90+	Chromium-based; full support
Safari (macOS/iOS)	14+	WebSocket and CSS Grid fully supported
Opera	76+	Full compatibility

Chapter 6

Database Design

6.1 Supabase Tables

The database schema consists of five primary tables. The `auth.users` table is managed internally by Supabase. Application tables reference user IDs from this table via foreign keys, with RLS policies ensuring data isolation.

Table: profiles

Extended user profile data created automatically via a database trigger on user registration.

Column	Type	Constraints	Description
<code>id</code>	UUID	PK, FK → <code>auth.users</code>	User identifier (matches <code>auth</code>)
<code>full_name</code>	TEXT	NOT NULL	Display name
<code>currency</code>	VARCHAR(5)	DEFAULT 'INR'	Preferred currency code
<code>created_at</code>	TIMESTAMPTZ	DEFAULT <code>now()</code>	Record creation timestamp
<code>updated_at</code>	TIMESTAMPTZ	DEFAULT <code>now()</code>	Last modification timestamp

Table: categories

Income and expense categories. System defaults have NULL `user_id`; user-defined categories reference the owner's UUID.

Column	Type	Constraints	Description
<code>id</code>	UUID	PK, DEFAULT <code>gen_uuid()</code>	Category identifier
<code>user_id</code>	UUID	FK → <code>auth.users</code> , NULLABLE	NULL = system default category
<code>name</code>	TEXT	NOT NULL	Category name (e.g. 'Groceries')
<code>type</code>	VARCHAR(10)	CHECK IN ('income','expense')	Transaction direction
<code>icon</code>	TEXT	NULLABLE	Emoji or icon class identifier
<code>color</code>	VARCHAR(7)	NULLABLE	Hex colour for UI badge (#RRGGBB)

Table: transactions

Core table recording every income and expense entry.

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_uuid()	Transaction identifier
user_id	UUID	FK → auth.users, NOT NULL	Record owner
category_id	UUID	FK → categories	Linked spending category
amount	NUMERIC(12,2)	NOT NULL, CHECK > 0	Transaction value
type	VARCHAR(10)	CHECK IN ('income','expense')	Direction
description	TEXT	NULLABLE	Optional memo or note
transaction_date	DATE	NOT NULL	Date the transaction occurred
created_at	TIMESTAMPTZ	DEFAULT now()	Row insertion timestamp

Table: budgets

Monthly budget limits per category per user.

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_uuid()	Budget identifier
user_id	UUID	FK → auth.users, NOT NULL	Budget owner
category_id	UUID	FK → categories	Target category
amount	NUMERIC(12,2)	NOT NULL, CHECK > 0	Monthly limit
month	DATE	NOT NULL	First day of the budget month
UNIQUE	(user_id, category_id, month)		One budget per category/month

6.2 Authentication

Authentication is fully managed by Supabase Auth (GoTrue server). The complete authentication flow:

- **Step 1:** User submits email and password via the React login form.
- **Step 2:** Supabase Auth validates credentials against the hashed password in `auth.users`.
- **Step 3:** On success, a signed JWT access token (1-hour expiry) and refresh token (7-day expiry) are returned.
- **Step 4:** The Supabase JS client stores tokens in `localStorage` and attaches the JWT to all API requests.
- **Step 5:** PostgreSQL RLS policies evaluate `auth.uid()` against the `user_id` column before permitting data access.
- **Step 6:** The client SDK auto-refreshes the access token when it approaches expiry, maintaining seamless sessions.

6.3 Data Flow

End-to-end data flow for a transaction creation operation.

Step	Actor	Action
1	User (Browser)	Fills the Add Transaction form and clicks 'Save'.
2	React	Validates inputs via React Hook Form; constructs the transaction payload.
3	Supabase JS SDK	Calls <code>supabase.from('transactions').insert({...})</code> with JWT in Authorization header.
4	PostgREST	Receives HTTPS POST; verifies JWT signature; forwards SQL INSERT to PostgreSQL.
5	PostgreSQL RLS	Evaluates INSERT policy: <code>auth.uid() = user_id</code> . Permits the operation.
6	PostgreSQL	Inserts row; WAL (Write-Ahead Log) records the change event.
7	Supabase Realtime	Broadcasts the INSERT event over WebSocket to all subscribed clients.
8	React	Receives the real-time payload; updates local state; re-renders the UI.

Chapter 7

System Architecture

The system follows a three-tier Jamstack architecture with clear separation between the presentation tier (React SPA on Netlify CDN), the logic/API tier (Supabase PostgREST and Auth), and the data tier (managed PostgreSQL).

7.1 High-Level Architecture

The diagram below illustrates the main subsystems and their interactions. The browser loads the pre-compiled React bundle from Netlify's CDN. Supabase Auth issues a JWT on login. Subsequent operations use this token with the PostgREST API, and real-time changes propagate via WebSocket.

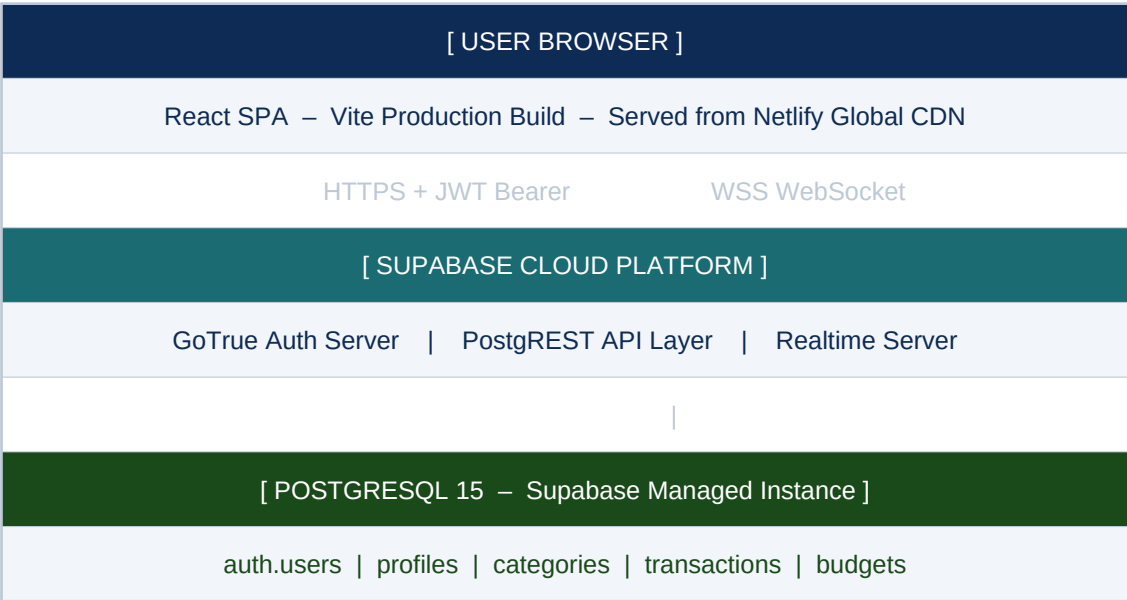


Figure 7.1 – High-Level System Architecture

7.2 React Component Hierarchy

```
App (Root – AuthProvider + BrowserRouter)
  ■■■ PublicRoutes: LoginPage | RegisterPage
  ■■■ PrivateRoutes: (requires valid JWT)
    ■■■ Layout (Navbar + Sidebar + Outlet)
      ■■■ DashboardPage → SummaryCards, IncomeExpenseBarChart, CategoryPieChart
```

```
■■■ TransactionsPage → TransactionList, AddTransactionModal, FilterBar
■■■ BudgetsPage → BudgetCard, BudgetProgressBar, AddBudgetModal
■■■ SettingsPage → ProfileForm, CategoryManager, ExportButton
```

Figure 7.2 – React Component Hierarchy

7.3 Security Architecture

Transport Layer: HTTPS enforced by Netlify using TLS 1.3. All Supabase API calls use encrypted HTTPS.

Authentication: JWT access tokens with 1-hour expiry issued by Supabase Auth; auto-refreshed via refresh tokens.

Authorisation: PostgreSQL RLS policies enforce `auth.uid() = user_id` on every table for SELECT, INSERT, UPDATE, DELETE.

Input Validation: React Hook Form validates client-side; PostgreSQL CHECK constraints enforce server-side rules.

Secret Management: Supabase URL and anon key stored as Netlify environment variables; never committed to version control.

CORS Policy: Supabase API configured to accept requests only from the registered Netlify production domain.

Chapter 8

Results and Output Screens

This chapter presents the key functional screens of the Personal Finance Tracker application as observed during testing and validation.

8.1 Login / Registration Screen

The authentication screen presents a card-based UI with email and password fields. Inline validation highlights errors before submission. A toggle between 'Login' and 'Sign Up' tabs supports new user registration with email verification via Supabase Auth. On success, the JWT is cached and the user is redirected to the Dashboard.

[Screenshot: Login / Registration Screen]

Figure 8.1 – Login / Registration Screen

8.2 Dashboard – Summary Overview

The main dashboard displays a three-card summary row (Total Income, Total Expenses, Net Savings) for the current month, updated in real-time. Below, a Recharts BarChart compares income vs. expenses by week, and a PieChart breaks spending into colour-coded categories. All cards update automatically when transactions are added.

[Screenshot: Dashboard – Income vs. Expense Summary]

Figure 8.2 – Dashboard Summary with Charts

8.3 Add Transaction Modal

Clicking the '+' FAB opens a modal. The user selects transaction type (income/expense), enters an amount, picks a date from a calendar picker, selects a category, and optionally adds a description. React Hook Form validates all required fields. A real-time subscription broadcasts the new row to other open sessions.

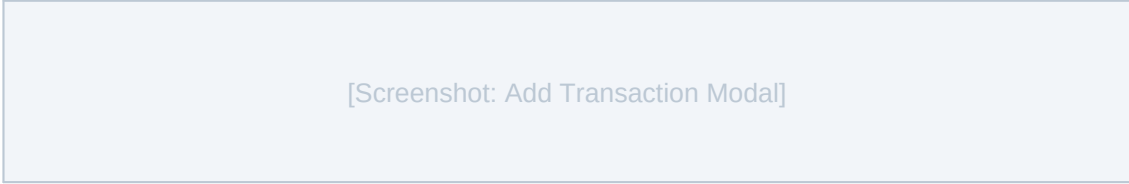
[Screenshot: Add Transaction Modal]

Figure 8.3 – Add Transaction Modal

8.4 Transaction History with Filters

The Transactions page renders a paginated list sorted by date (newest first). Filter controls allow narrowing by date range, category, and transaction type. Each row shows the category icon, description, date, and amount (green for income, red for expense). Row-level delete with confirmation ensures no accidental removal.

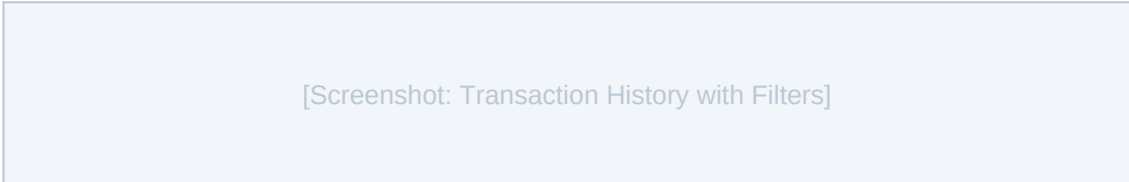
[Screenshot: Transaction History with Filters]

Figure 8.4 – Transaction History with Filters

8.5 Budget Management Screen

The Budgets page lists all monthly budgets as progress cards. Each card shows the category, budget limit, amount spent, and a colour-coded progress bar: green below 75%, amber from 75–90%, red above 90%. An 'Add Budget' button opens a form for any category and month combination. Overspent categories display a warning badge.

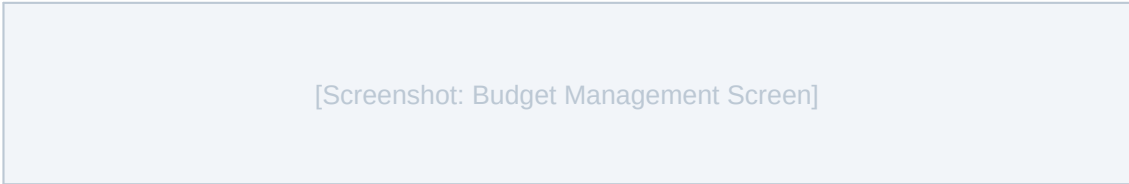
[Screenshot: Budget Management Screen]

Figure 8.5 – Budget Management Screen

8.6 CSV Export Feature

The Settings page provides an Export section where users select a date range and click 'Export CSV'. PapaParse serialises all matching transactions into a downloadable file with columns: Date, Type, Category, Description, Amount. The file opens natively in Microsoft Excel or Google Sheets for further analysis.

[Screenshot: CSV Export Feature]

Figure 8.6 – CSV Export

6.7 Performance Testing Results

Metric	Result	Target
Lighthouse Performance Score	92 / 100	≥ 90
First Contentful Paint (FCP)	0.8 s	< 1.5 s
Time to Interactive (TTI)	1.4 s	< 3.0 s
Largest Contentful Paint (LCP)	1.2 s	< 2.5 s
Cumulative Layout Shift (CLS)	0.02	< 0.1
Supabase API Response Time (avg)	< 120 ms	< 300 ms
End-to-End Transaction Insert	< 200 ms	< 500 ms

Chapter 9

Conclusion and Future Scope

9.1 Conclusion

This internship project successfully demonstrates the end-to-end design and development of a Personal Finance Tracker Web Application using a modern Jamstack technology stack. All primary objectives were achieved:

- A responsive, component-based React frontend was built and deployed as a high-performance SPA on Netlify.
- Secure user authentication was implemented with Supabase Auth and JWT-based session management.
- A normalised PostgreSQL schema with Row-Level Security ensures complete data isolation between users.
- Real-time transaction updates are delivered via Supabase WebSocket subscriptions without page refreshes.
- CI/CD deployment from GitHub to Netlify enables rapid, automated production releases.
- Performance testing confirmed Lighthouse scores above 90 and sub-200 ms end-to-end operation times.

The internship provided invaluable hands-on exposure to the complete software development lifecycle – from eliciting requirements and designing architecture through implementing features, writing tests, and deploying to production. Working with a modern BaaS platform accelerated development timelines while maintaining production-grade security and performance standards.

9.2 Future Scope

Multi-Currency Support: Integrate with an open exchange rates API to support transactions in multiple currencies with automatic conversion to a user-selected base currency.

Open Banking Integration: Use Plaid (international) or Finvu (India) APIs to automatically import bank and card transactions, eliminating manual data entry.

AI-Powered Insights: Integrate a language model API to analyse spending patterns and provide personalised, contextual saving recommendations.

Recurring Transaction Engine: Allow users to define recurring income (salary, dividends) or expense (rent, subscriptions) entries that auto-populate monthly.

Shared Household Budgets: Implement multi-user shared workspaces for couples or families with role-based access control.

Mobile App (React Native): Port the UI to React Native sharing the same business logic layer with the web version for native iOS and Android apps.

Bill Reminders: Email and push notifications via Supabase Edge Functions and a transactional email provider when bill due dates approach.

Tax Report Generation: Auto-generate annual income and expense summaries formatted to assist with Indian ITR filing.

Receipt OCR: Use Tesseract.js or a cloud vision API to extract transaction amounts and dates from photographed receipts.

Offline PWA Mode: Implement a Service Worker with IndexedDB for offline transaction entry, syncing with Supabase when connectivity resumes.

The project's component-based frontend, serverless backend, and managed database service are all horizontally scalable, making the platform suitable for a growing user base without significant re-architecture.

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